



# ETHICS AND LAW IN DATA AND ANALYTICS

Understanding the Ethical  
and Legal Frameworks

An abstract network diagram with several grey cylindrical nodes of varying sizes connected by thin grey lines. The nodes are arranged in a non-uniform pattern across the slide. A vertical blue line is positioned to the right of the word 'INTRODUCTION'.

# INTRODUCTION

Data ethics refers to the principles behind how organizations gather, protect, and use data. It's a field of ethics that focuses on the moral obligations that entities have (or should have) when collecting and disseminating information about us.

Article review:

<https://www.datacamp.com/blog/introduction-to-data-ethics>

# 5 C'S

- 1). Consent (making sure you have an agreement when obtaining consent to collect and use data);
- 2). Clarity (being clear about what users are consenting to);
- 3). Consistency & trust (being predictable and trustworthy in intentions and actions);
- 4). Control & transparency (being clear about what controls users have over their own data and what to do if they change their mind); and
- 5). Consequences (thinking through potential problems or issues before they occur).

<https://dataethicsrepository.iaa.ncsu.edu/2023/04/18/the-five-cs-five-framing-guidelines-to-help-you-think-about-building-data-products/>

## KEY PRINCIPLES:

- Ownership

- Transparency

- Privacy

- Intention

- Outcomes

# OWNERSHIP

## 1. Ownership

**Description:** This principle emphasizes that individuals should have control over their own data. They should be aware of what data is being collected, how it is being used, and have the ability to consent to or deny its use.

**Importance:** Ensures respect for personal autonomy and privacy.

**Examples:** User consent forms, data access requests.



# TRANSPARENCY

## 2. Transparency

**Description:** Organizations should be open about their data practices. This includes being clear about what data is collected, how it is used, and who it is shared with.

**Importance:** Builds trust between organizations and individuals.

**Examples:** Clear privacy policies, transparent data usage statements.



# PRIVACY

## 3. Privacy

**Description:** Protecting individuals' data from unauthorized access and ensuring that personal information is kept confidential.

**Importance:** Safeguards personal information and prevents misuse.

**Examples:** Data anonymization, encryption, secure data storage.



# INTENTION

## 4. Intention

**Description:** Data should be collected and used with a clear, ethical purpose in mind. The intentions behind data collection should align with the values and expectations of the individuals whose data is being used.

**Importance:** Prevents misuse of data and ensures ethical alignment.

**Examples:** Using data for improving services rather than for exploitative purposes.





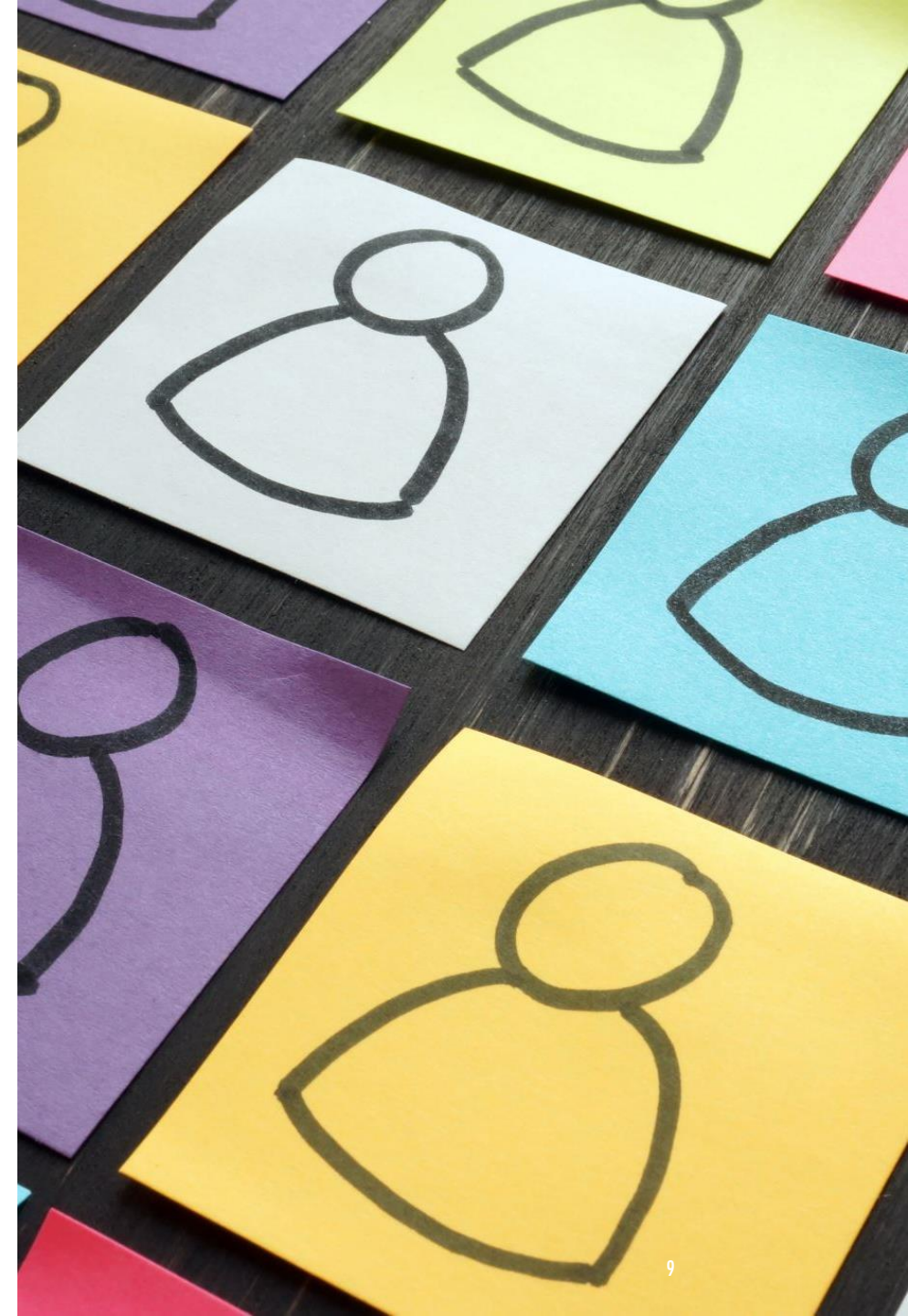
# OUTCOMES

## 5. Outcomes

**Description:** Considering the potential impacts and consequences of data use on individuals and society. This includes assessing both positive and negative outcomes.

**Importance:** Ensures that data practices contribute positively to society and do not cause harm.

**Examples:** Conducting impact assessments, monitoring for unintended consequences.



# IMPORTANCE OF DATA ETHICS

- **Trust:** Ethical data practices build trust between organizations and individuals, fostering a positive relationship.
- **Compliance:** Adhering to data ethics helps organizations comply with legal and regulatory requirements.
- **Reputation:** Ethical data practices enhance an organization's reputation and credibility.
- **Innovation:** Promotes responsible innovation by ensuring that new technologies and data uses are developed with ethical considerations in mind.

# VIDEO

<https://www.youtube.com/watch?v=woUFrmrJVh4&t=139s>

# CASE ANALYSIS

<https://fotislaw.com/public/lawtify/case-study-on-facebooks-data-breach>

Create an affinity diagram to analyze this case study.



# DATA ETHICS EXAMPLES

<https://atlan.com/data-ethics-examples/>



# DATA ETHICS- VIDEO

<https://www.youtube.com/watch?v=LQeSExnIDtw&t=205s>



# THE DATA REVOLUTION

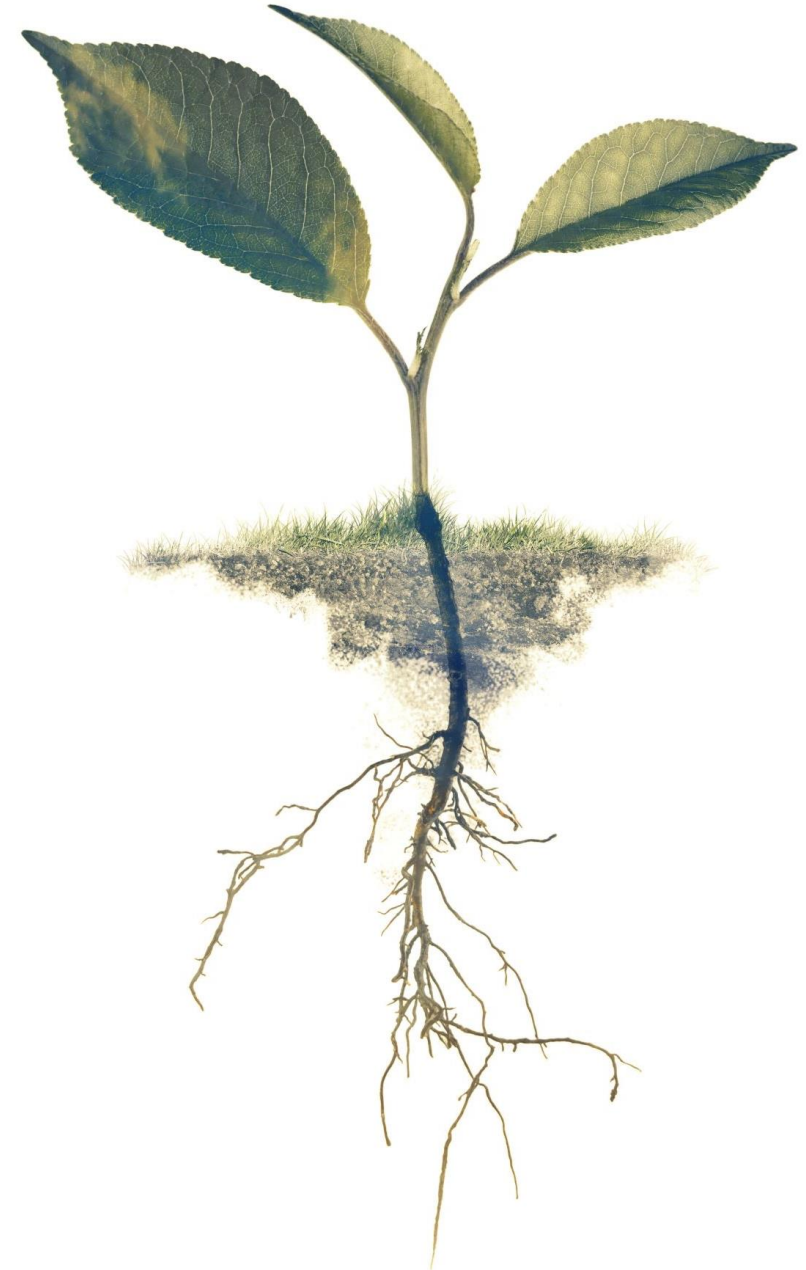
Growth of big data and AI

Impact on society and businesses

Ethical considerations

# GROWTH OF DIGITAL ERA

[https://iabaustralia.com.au/wp-content/uploads/2020/04/1493349702\\_b8633d46c49a96bfe0eb94f9cb3d9800.jpg](https://iabaustralia.com.au/wp-content/uploads/2020/04/1493349702_b8633d46c49a96bfe0eb94f9cb3d9800.jpg)



# HISTORY

<https://medium.com/@vamshikrish2015/the-evolution-of-data-engineering-a-walk-through-time-1b559f17a59>

<https://www.techtarget.com/whatis/feature/A-history-and-timeline-of-big-data>

# LEGAL FRAMEWORKS

General Data Protection Regulation (GDPR)

California Consumer Privacy Act (CCPA)

Other relevant laws and regulations



# GDPR

The General Data Protection Regulation (GDPR) is a legal framework that sets guidelines for the collection and processing of personal information from individuals who live in and outside of the European Union (EU).

<https://www.investopedia.com/terms/g/general-data-protection-regulation-gdpr.asp>





# VIDEO

[youtube.com/watch?v=MJ\\_Qj4H9ixc](https://youtube.com/watch?v=MJ_Qj4H9ixc)

# CCPA

The California Consumer Privacy Act of 2018 (CCPA) gives consumers more control over the personal information that businesses collect about them and the CCPA regulations provide guidance on how to implement the law.

<https://www.oag.ca.gov/privacy/ccpa>



# VIDEO

<https://www.youtube.com/watch?v=dpzsAgrDAO4>

The background of the slide is a dark, high-contrast image. It features a close-up of a computer keyboard on the left side, with keys like 'F', 'G', 'H', and 'J' visible. The right side of the image shows a blurred view of electronic circuitry, possibly a motherboard, with various components and traces. The overall tone is dark and technological.

# OVERVIEW OF GDPR, CCPA, AND OTHER RELEVANT LAWS AND REGULATIONS

<https://www.youtube.com/watch?v=Xl5r5JWqclU&t=231s>

# RESEARCH

Divide in groups.

Create a Powerpoint on -What is CPRA?

*Do not use AI for creating PPT; start from scratch; use your own research.*



# REPORT

[https://www.dataguidance.com/sites/default/files/20220127\\_onetrust\\_dataguidance\\_gdpr\\_v\\_ccpa\\_and\\_cpri\\_report.pdf](https://www.dataguidance.com/sites/default/files/20220127_onetrust_dataguidance_gdpr_v_ccpa_and_cpri_report.pdf)





# DATA PRIVACY AND SECURITY

Importance of data privacy

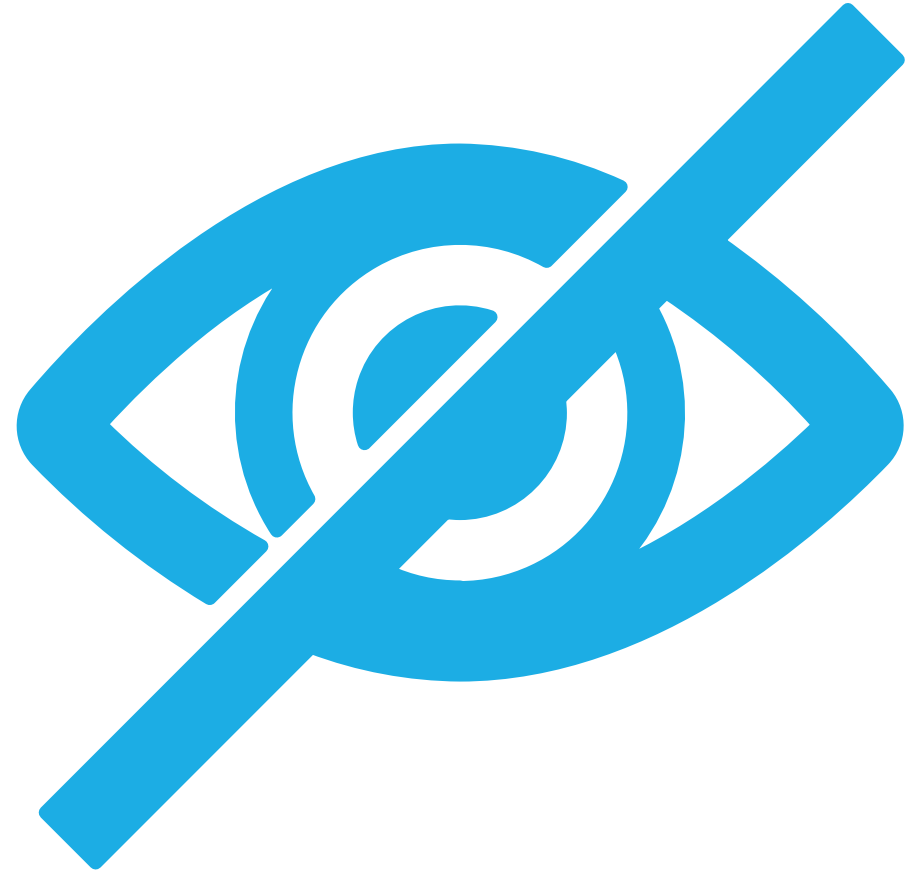
Data breaches and their consequences

Best practices for data security

# DATA PRIVACY

Data privacy, also called "information privacy," is the principle that a person should have control over their personal data, including the ability to decide how organizations collect, store and use their data.

<https://www.ibm.com/think/topics/data-privacy>



# UBER BREACH CASE STUDY



<https://humanfirewall.io/the-uber-breach-case-study-cybersecurity-lessons-learned/>



Do a cause effect analysis on the case study; use x-mind.

# BEST PRACTICES

[https://www.youtube.com/watch?v=jq\\_LZ1RFPfU&t=846s](https://www.youtube.com/watch?v=jq_LZ1RFPfU&t=846s)

# BIAS AND FAIRNESS IN DATA ANALYTICS

Types of bias in data processing

Impact of bias on decision-making

Strategies to mitigate bias



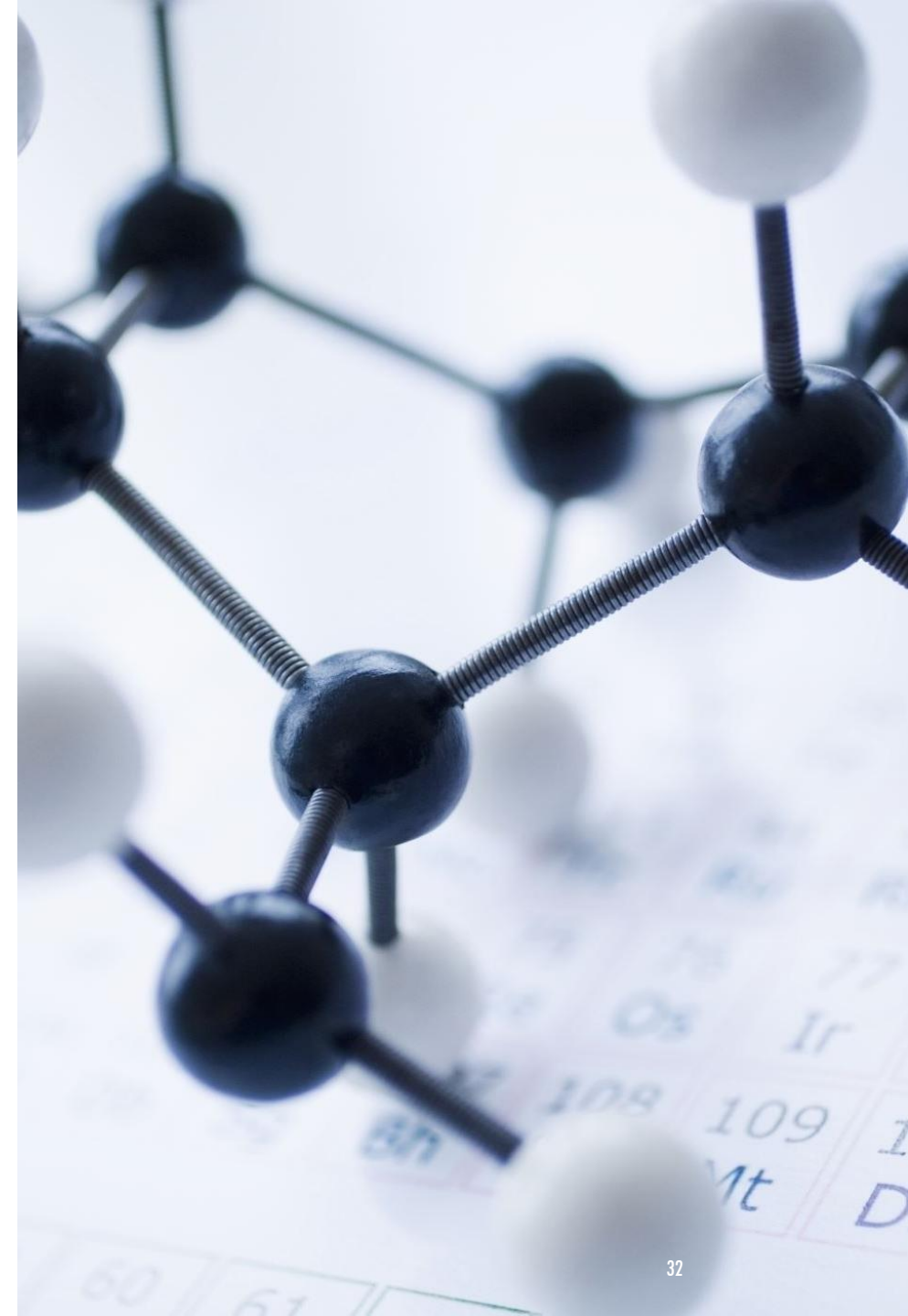
# TYPES OF BIAS IN DATA

<https://towardsdatascience.com/types-of-biases-in-data-cafc4f2634fb>

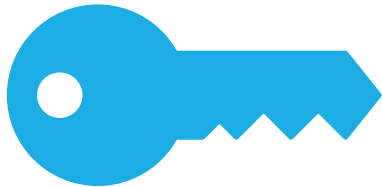


# MITIGATING BIAS

<https://medium.com/@marquesjef/understanding-and-mitigating-data-bias-in-data-analysis-and-data-science-15a5e9282356>



# CONCLUSION



Summary: Recap of key points



Call to Action: Encouraging ethical practices in data analytics